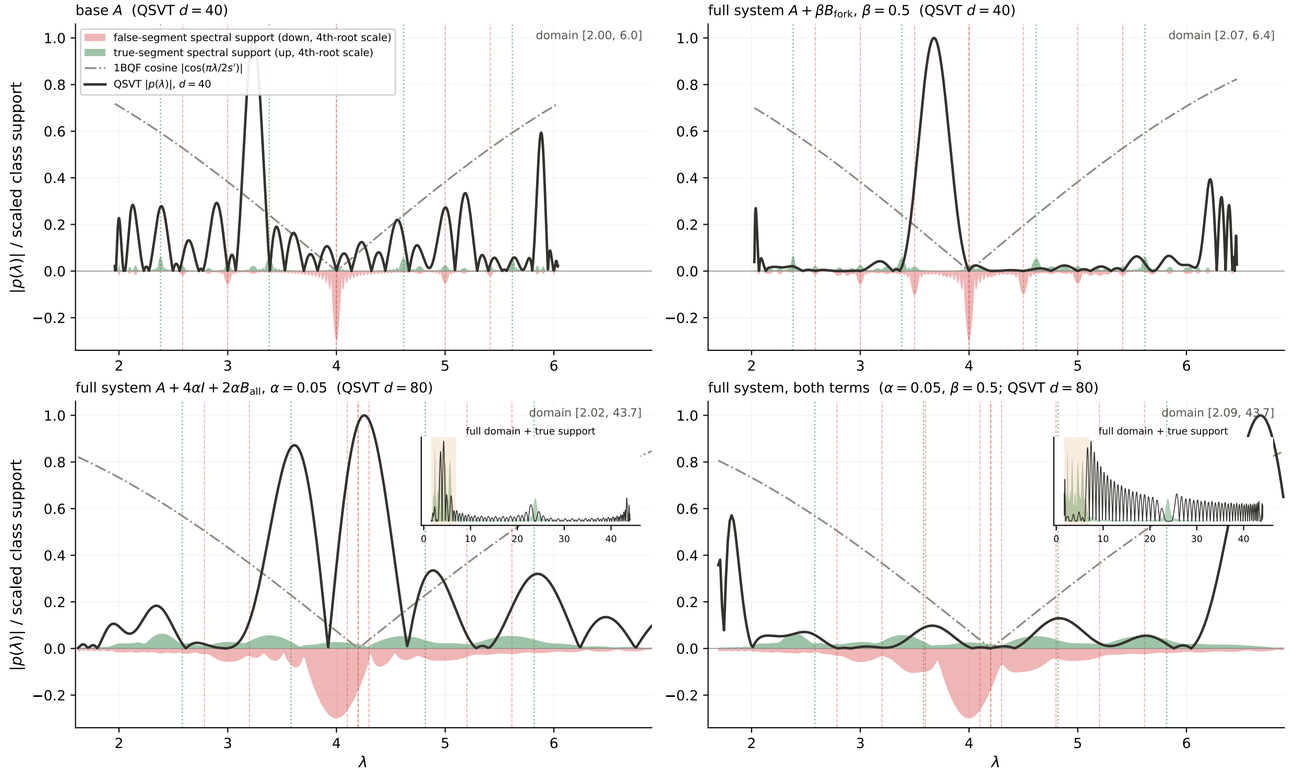
The QSVT polynomial (refit per Hamiltonian) against each Hamiltonian's MEASURED class spectral support (moderate configuration, $T = 200$, rep 0)



shaded: class-projected densities of states $D_C(\lambda) = \sum_{i \in C} \sum_j v_j(i)^2\, \delta(\lambda - \lambda_j)$ (Hutchinson + Chebyshev-KPM, Jackson kernel, $K = 600$; true drawn up, false drawn down, 4th-root scale) · dotted/dashed verticals: closed-form isolated-motif lines (vocabulary, not population) · every operator is the FULL system, base A + added term(s) · $\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 10\,\mu$m, drop 1%, formula $\varepsilon = 3.18$ mrad, $\gamma = 3$, $\delta = 1$ · drawn responses reproduce the study CSV numbers to 10^{-9}